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Evolution of the North Qiangtang Block in the late Paleozoic: Paleomagnetism and its tectonic implications

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ABSTRACT

Understanding of the geodynamic evolution of the Tethyan realm cannot be complete without paleogeographical reconstructions of the North Qiangtang Block (NQB) that occupies a central position in Tibet. However, the reliability of such a reconstruction for the Paleozoic still requires substantial improvement. In this paper, we present paleomagnetic results obtained from the Middle Permian limestones and Upper Permian volcanic rocks in the Tanggula area, aiming to provide precise constraints on the NQB kinematics during the geodynamic evolution of the Tethys realm. Combined with other available paleomagnetic data from the NQB, our results suggest that the block was stably located at ~24°S for a long time before the Middle Permian, started to drift rapidly northward in the Middle Permian, and reached ~8.4°S in the Late Permian. The NQB continued drifting rapidly northward during the Triassic until merging with the southern margin of Eurasia in the Late Triassic. We reviewed new and available paleomagnetic and geological data and proposed a revised model for the tectonic evolution of the eastern Tethys realm. The NQB likely belonged to a ribbon-like separate continental archipelago in the middle of the Paleo-Tethys Ocean during the early–late Paleozoic. The Cimmerian continent, including the South Qiangtang Block (SQB), rifted away from the northern margin of Gondwana and drifted northward to ~22°S in the Middle Permian. The continent approached or partially collided with the NQB, resulting in the rapid northward movement of the NQB and the formation of the Longmuco–Shuanghu suture between the NQB and SQB.

INTRODUCTION

The Qinghai-Tibet Plateau in the eastern Tethys realm is composed of several continental blocks that originated from the Gondwana or Laurasia continents and their surroundings (Yin and Harrison, 2000; Pan et al., 2012; Metcalfe, 2013; Zhu et al., 2013; Huang et al., 2018; Zhu et al., 2022). The Qiangtang area, which mainly consists of the South and North Qiangtang blocks (SQB and NQB), is located at the hinterland of the Qinghai-Tibet Plateau (Fig. 1; e.g., Li et al., 2009; Pan et al., 2012). These blocks were traditionally regarded as a part of the Cimmerian continent introduced by Şengör (1979, 1984). The continent consisted of several Gondwana-originated blocks that rifted away from the northern margin of Gondwana in the Carboniferous–Early Permian. During the latest Paleozoic and early Mesozoic, the continent drifted northward across the Tethys Ocean and finally amalgamated with the southern part of the Laurasia continent in the Late Triassic. It led to the closure of the Paleo-Tethys Ocean and the opening of the Neo-Tethys Ocean (Şengör, 1979, 1984; Metcalfe, 1991, 2013; Şengör and Atayman, 2009; Yin and Harrison, 2000; Torsvik and Cocks, 2013; Zhu et al., 2013; Wang et al., 2018; Huang et al., 2018; Wu et al., 2020; Zhu et al., 2022). Therefore, reconstructing the late Paleozoic paleogeography of the NQB and SQB and their affinities with the Gondwana or Laurasia continents is crucial in qualifying the Tethys Ocean evolution in the East Tethys domain.

Paleomagnetism is an essential tool for reconstructing the paleogeographic and kinematic processes of tectonic blocks. A series of paleomagnetic studies have been conducted in recent decades on the late Paleozoic and early Mesozoic strata of the SQB and NQB to constrain the paleogeography and provide critical controls on the drift history of these blocks (Fig. 1; e.g., Huang et al., 1992; Cheng et al., 2012a, 2012b, 2013; Song et al., 2012; Song et al., 2015, 2017, 2020; Chen et al., 2017; Yang et al., 2017; Ma et al., 2019; Zhou et al., 2019; Cao et al., 2019, 2020; Guan et al., 2021; Jiang et al., 2022; Yu et al., 2022; Wei et al., 2022). Available paleomagnetic data show that the SQB was located at ~22°S in the Middle Permian (Wei et al., 2022) and merged with the NQB before the Late Triassic (Song et al., 2012). The NQB was similarly located at low to mid-latitudes in the southern hemisphere during the late Paleozoic (e.g., Cheng et al., 2012b, 2013; Yang et al., 2017; Song et al., 2017), drifted rapidly northward during the Early and Late Triassic (Zhou et al., 2019), and merged into the southern margin of the North China and Tarim blocks in the Late Triassic (e.g., Song et al., 2015; Zhou et al., 2019; Yu et al., 2022). Therefore, the paleomagnetic data appear to reveal an understanding similar to the conventional view. Both the NQB and SQB originated in Gondwana, splitting from the northern margin of Gondwana successively in the Carboniferous–Permian and drifting rapidly northward across the Tethys Ocean, transforming the Paleo-Tethys Ocean into the Neo-Tethys Ocean. However, this view is discordant with new geological evidence from the Qiangtang area (e.g., Li, 1987; Zhang et al., 2006a, 2006b; Zhai et al., 2011; Tang and Zhang, 2014; Zhang et al., 2016; Liu et al., 2019). The Paleozoic stratigraphic sequences of the NQB present pro-Laurasia elements, indicating that the block may have a closer affinity with Laurasia. For instance, the correlation between the metamorphic basement and the early Paleozoic strata with the contemporaneous Laurasia-affinity blocks (Wu et al., 2020), the angular unconformity between the Devonian and pre-Devonian strata indicating the influence of the Caledonian movement (Li

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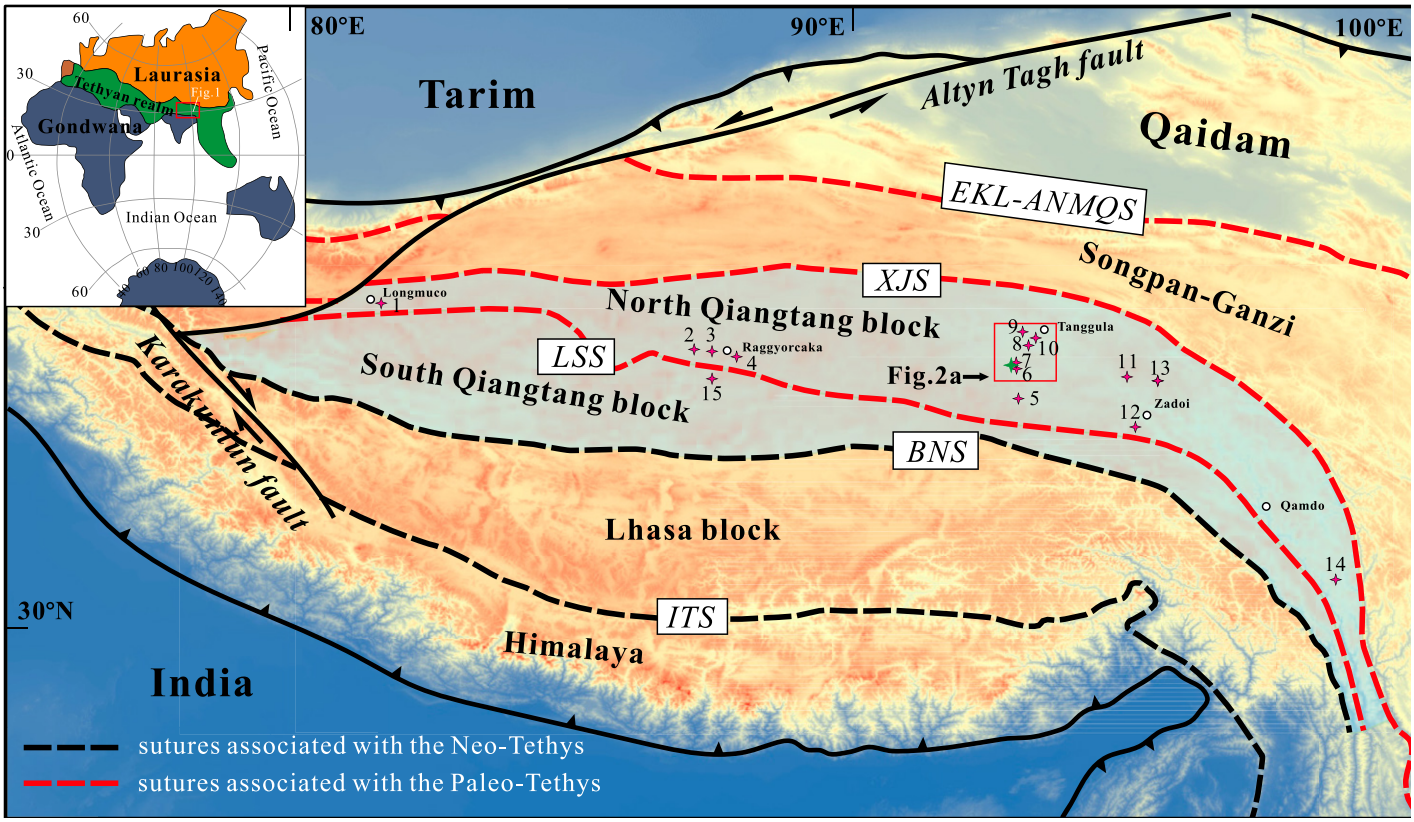


Figure 1. The tectonic framework of the Qinghai-Tibet Plateau shows the locations of the major Tibetan blocks, suture zones, and our study area. The red crosses indicate locations of the Paleozoic and Mesozoic paleomagnetic studies in the Qiangtang area: 1—Jiang et al. (2022); 2—Cheng et al. (2012b); 3—Zhou et al. (2019); 4—Song et al. (2012); 5—Song et al. (2020); 6—Cheng et al. (2013); 7—Ma et al. (2019); 8—Yang et al. (2017); 9—Zhou et al. (2019); 10—Song et al. (2015); 11—Yu et al. (2022); 12—Guan et al. (2021); 13—Lin and Watts (1988); 14—Huang et al. (1992); 15—Wei et al. (2022). The green cross indicates this study. BNS—Bangong Co–Nujiang Suture Zone; EKL-ANMQS—Eastern Kunlun–Animaqing Suture Zone; ITS—Indus–Yarlung Zangbo Suture Zone; LSS—Longmuco–Shuanghu Suture Zone; XJS—Xijinwulan-Jinshajiang Suture Zone.

et al., 2006), and the distinct Permo-Carboniferous warm-water flora and fauna (Zhang et al., 2013) predict that the Paleozoic evolution of the NQB is pretty different from that of Gondwana affinity blocks (e.g., SQB and Lhasa). These Gondwana affinity blocks are notably characterized by the Pan-African crystalline basement (Li et al., 2006). The development of glacial deposits and the appearance of cold-water biota accompanied the development of Gondwana-type Carboniferous–Permian sedimentary rocks (Zhang et al., 2013, 2016). An angularly unconformable contact is observed between the Ordovician and the underlying Cambrian and Precambrian (Li et al., 2006; Wu et al., 2020). The new geological discoveries also suggest that the Longmuco–Shuanghu Suture Zone (LSS), located between the NQB and SQB, likely represents the remains of the main oceanic branch of the Paleo-Tethys Ocean (e.g., Li et al., 2006; Zhai et al., 2013, 2016). The branch is called the Longmuco-Shuanghu Ocean (LSO) in the literature (Huang et al., 2018). There are several E-W–trending Tethys suture zones in the Qinghai-Tibet Plateau (Fig. 1), including the LSS, the Xijinwulan-Jinshajiang Suture Zone (XJS), and the Eastern Kunlun–Animaqing Suture Zone (EKL-ANMQS). These zones are associated with the Paleo-Tethys, and the Bangong Co–Nujiang and India–Yarlung Zangbo Suture zones relate to the Neo-Tethys (Huang et al., 2018; Wang et al., 2018; Wu et al., 2020; Zhu et al., 2022). The LSS in central Qiangtang is a 600-km-long metamorphic belt consisting mainly of Cambrian to Triassic ophiolitic mélanges (Zhai et al., 2013, 2016) and Permian to Triassic high-pressure/low-temperature metamorphic rocks (Kapp et al., 2000; Pullen et al., 2008). Li (1987) proposed the LSS as a first-order plate boundary between Paleozoic Gondwana and Laurasia, representing the in situ relic of the main branch of the Paleo-Tethys Ocean. The accumulating geological evidence supports this suggestion (e.g., Metcalfe, 1991, 2013; Li et al., 2006; Zhai et al., 2013, 2016; Liu et al., 2022). At the same time, there exist suggestions that there is no suture in the middle of Qiangtang and that the central Qiangtang metamorphic belt was formed during the Middle Triassic as a result of the XJS southward subduction of the northern edge of the Qiangtang Block (Kapp et al., 2000; Pullen et al., 2008; Pullen and Kapp, 2014). Conclusively, the uncertainty about the affinities between the NQB and Gondawana or Laurasia has led to confusion about the nature and evolutionary history of the Tethys Ocean and Qinghai-Tibet Plateau suture zones. Our study aims to obtain new high-quality paleomagnetic data to resolve this uncertainty.

In this paper, we report new Middle and Late Permian paleomagnetic data from the Tanggula area of eastern Tibet. Our study aims to provide high-precision constraints on the paleogeography and drift history of the NQB. We use our new paleomagnetic data to revise the plate tectonic evolution of the eastern part of the Tethys Ocean.

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GEOLOGICAL SETTING AND PALEOMAGNETIC SAMPLING

The late Paleozoic strata of the NQB mainly distribute in the Qamdo, Zadoi, Tanggula, Raggyorcaka, and Longmuco areas (Fig. 1; Zhang et al., 2013). We collected samples in the Tanggula area, next to the Qinghai-Tibet Railway between the Tuotuo and Tongtian rivers.

The late Paleozoic strata exposed in the Tanggula area include the Upper Carboniferous Zharigen Formation (C_2z), the Lower Permian Nuoribagaribao Formation (P_1nr), the Middle Permian Jiushidaoban Formation (P_2j), the Upper Permian Nayixiong Formation (P_3n), and the Upper Permian Labuzhari Formation (P_3lb), which are coal-bearing clastic rocks and carbonate intercalated volcanic rocks (Fig. 2; Liu and Tian, 1993; Niu et al., 2010). The Upper Triassic or Middle Jurassic overlies the late Paleozoic with an angular unconformity (Geological Survey of Qinghai Province, 2005; Geological Survey of Chengdu University of Technology, 2005). The Upper Triassic strata include the Jiapila Formation (T_3jp), Bolila Formation (T_3b), and Bagong Formation (T_3bg), which are mainly composed of clastic and carbonate rocks with few volcanic rocks, and T_3b contains coal. The Middle Jurassic strata include the Quemocuo Formation (J_2q) and the Buqu Formation (J_2b), which are composed of clastic rocks and limestone.

We collected paleomagnetic samples from two sections (section A–A′ and section B–B′) situated in two opposite wings of the Zhouqiongmalu compound anticline (Fig. 2A). The structure consists of P_1nr, P_2j, and P_3n, extending in the direction of NW-SE. The anticline strata are mainly Permian, with the sporadically exposed Triassic rocks as fault blocks. The Jurassic sediments unconformably cover the anticline. These features indicate that the anticline was formed before the Jurassic Period.

Section A–A′ is the volcanic outcrop of P_3n. The bedding dips to the northeast at an angle of ~15°–45°. Section A–A′ can be divided into three parts (Figs. 3A–3G). The upper part is ~5 m thick and consists mainly of rhyolite and andesite. The rhyolite is light purple and has a porphyritic texture and rhyolite structure when observed under a microscope. The phenocrysts are composed of quartz and alkali feldspar. The basalt is reddish brown with a porphyritic texture and massive structure. The middle part is ~35 m thick and consists mainly of moderately thick layered tuffs with welded breccias at the bottom. The fresh surface of the tuff is gray-green with a clastic tuff structure and massive structure. The rocks consist mainly of crystal chips and volcanic ash. The lower part is ~30 m thick and consists primarily of andesite and basalt. Basalt hand specimens are purplish-red and have a porphyritic texture under a microscope. The phenocrysts are plagioclase, pyroxene, and calcite, with occasional carbonate and chlorite filled in the stomata. Andesite hand specimens are purplish-red and reddish-brown with porphyritic texture and massive structure; phenocrysts contain plagioclase and hornblende. Volcanic rocks characterize by multiple cycles and contain multiple sedimentary interlayers. We dredged two samples (B01 and B02) from the upper and lower parts of section A–A′ for detailed petrological, geochemical, and chronological analyses (Zhang et al., 2019). The zircon U-Pb dating results show an age of 249.6 ± 1.3 Ma for B01 and 251.1 ± 4.8 Ma for B02, indicating that the volcanic rocks are ca. 250 Ma. Based

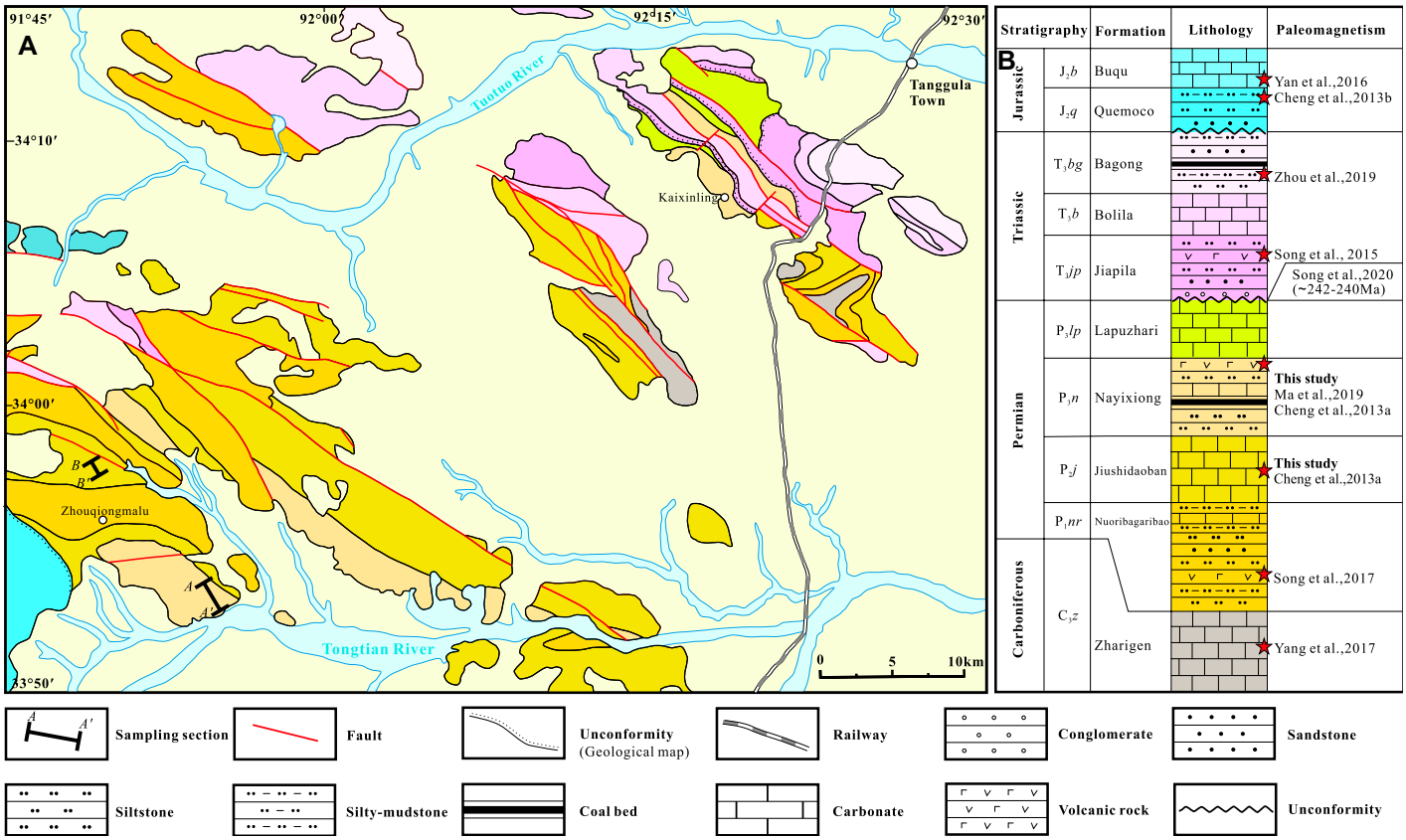


Figure 2. Sketches of geology and sampling location for this study in eastern Tibet. (A) Simplified geological map and (B) schematic lithostratigraphic column of the Tanggula area (modified from Geological Survey of Qinghai Province, 2005; Geological Survey of Chengdu University of Technology, 2005; Liu et al., 2016).

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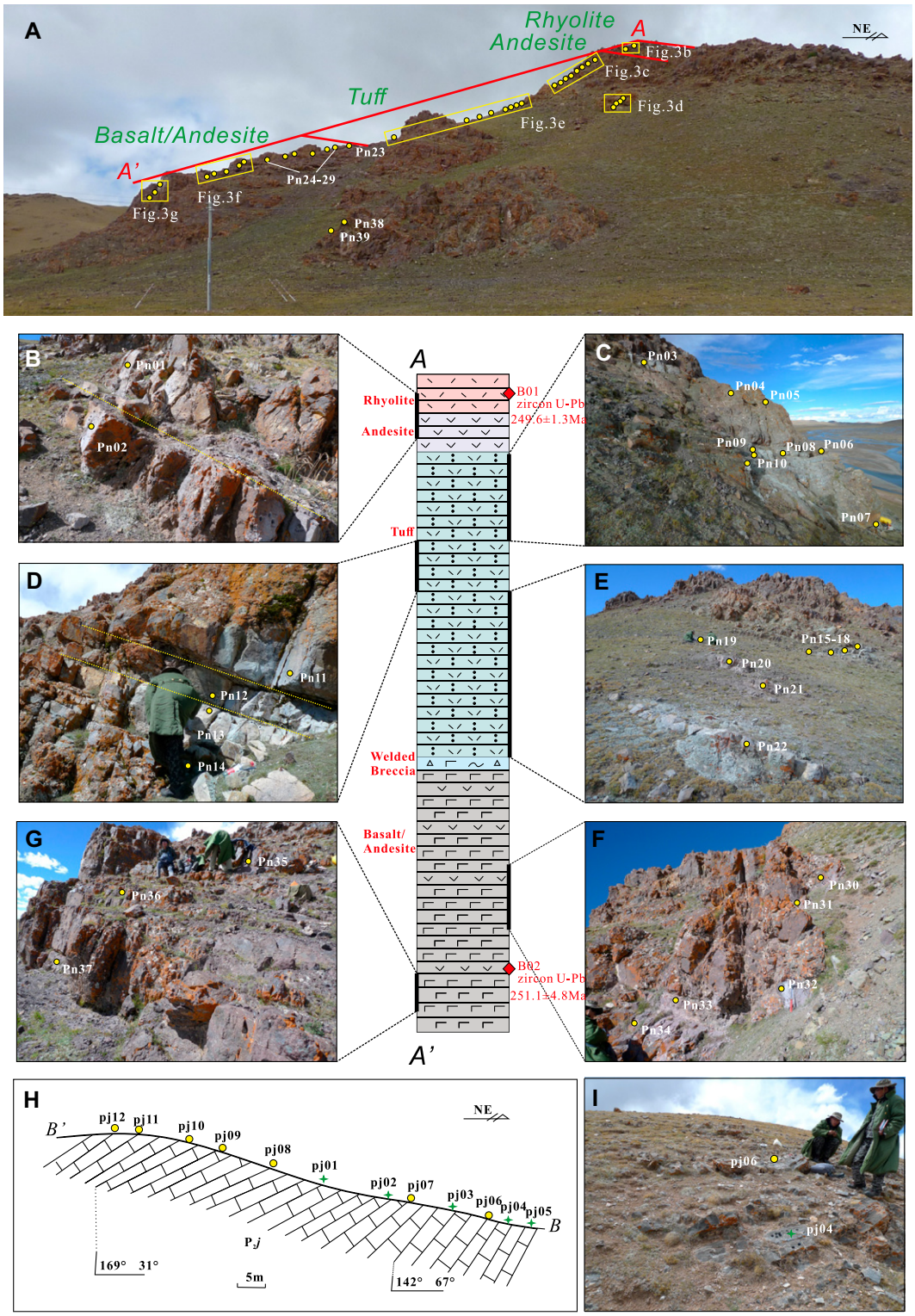


Figure 3. (A–G) Photographs showing field outcrops and sampling sites in section A–A′ for this study in eastern Tibet. (H) Simplified geological section and (I) a photograph showing the sampling of section B–B′. Yellow circles indicate sites in this study; green crosses indicate sites from Cheng et al. (2013); red squares in the schematic lithostratigraphic column of section A–A′ indicate the locations of the zircon U-Pb dating samples from Zhang et al. (2019).

on the geochemical composition of the volcanic rocks, they probably formed in an island-arc tectonic setting.

We sampled paleomagnetic sites layer-by-layer in section A–A′ (Figs. 3A–3G), with one site (9 samples) of rhyolite and one site (10 samples) of andesite in the upper part, 20 sites (151 samples) of tuff layer by layer in the middle part, one site (8 samples) of welded breccia at the bottom of the tuff, and 16 sites (128 samples) of basalt and andesite in the lower part (each site from one or more lava flows). Finally, we collected 306 independently oriented paleomagnetic samples in 39 sites.

Section B–B′ exposes the limestone of P_2j (Figs. 3H and 3I). The stratigraphy dips to the southwest at an angle of 30°–71°. It consists mainly of fossiliferous microcrystalline lime-

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stone, ting limestone, and coral reef limestone of the Maokou and Lengwu stage (middle-upper part of the Middle Permian) (Geological Survey of Chengdu University of Technology, 2005; Qiao et al., 2021).

We previously reported paleomagnetic data from five sites (42 samples) in section B–B′ (Cheng et al., 2013); this time, we collected samples from seven other sites (65 samples) (Figs. 3H and 3I). This study analyzes all 12 sites together to produce a statistically sound result.

The oriented drill core samples were collected using a portable gasoline-powered drill and oriented with magnetic and solar compasses. The cores were cut into 2.2-cm-long standard specimens for subsequent rock magnetic and paleomagnetic analysis in the laboratory.

ANALYTICAL METHODS

Rock-magnetic and paleomagnetic measurements were carried out in the laboratory of paleomagnetism and geochronology at the Institute of Geology and Geophysics, Chinese Academy of Sciences (Beijing, China), and the paleomagnetic laboratory of the State Key Laboratory of Continental Dynamics, Northwest University (Xi'an, China).

We used one representative sample per site to examine the rock-magnetic properties. The progressive acquisition of isothermal remanent magnetization (IRM) and subsequent stepwise thermal demagnetization of a three-axis IRM (Lowrie, 1990) were obtained by magnetizing samples in the field of 2700 mT, 400 mT, and 40 mT along the three orthogonal axes Z, Y, and X, respectively. The magnetic field was produced with the ASC IM-10-30 pulse magnetizer and measured with the AGICO JR-5A spinner magnetometer. The thermal demagnetization of the composite IRM was performed with an MMTD-48 furnace and a 2G-755R superconducting magnetometer.

The anisotropy of magnetic susceptibility (AMS) of the remaining samples was measured using the AGICO Kappabridge KLY-4S susceptibility meter. For the volcanic samples, the characteristic remanent magnetization (ChRM) was separated using stepwise thermal demagnetization; for the limestone, a hybrid demagnetization method was used, consisting of first progressive thermal demagnetization up to 250 °C followed by alternating field (AF) demagnetization to a peak field of ~100 mT. Progressive thermal demagnetization was performed using the MMTD-48 furnace in steps of 50–100 °C below 450 °C and 10–25 °C above 450 °C. A stepwise alternating demagnetization of 5–10 mT was performed using a 2G-600 alternating field demagnetizer. The remanent magnetization was measured on the 2G-755R low-temperature superconducting magnetometer. The demagnetization and remanence measurements were performed in a magnetically shielded space with a residual magnetic field of less than 300 nT in the center.

We projected the demagnetization results to the orthogonal vector diagram (Zijderveld, 1967) and the stereographic projection map. The principal component analysis (PCA) (Kirschvink, 1980) or converging remagnetization great circle (GC) method (Halls, 1976) was used to determine the remanence directions. We calculated mean site directions using the Fisher statistics (Fisher, 1953) or the method that combines the great-circle fits and ChRM directions (McFadden and McElhinny, 1988) via PMGSC (Enkin, 1994) and PaleoMac 6.5 (Cogné, 2003) software and performed paleogeographic reconstructions using the GPlates software (Müller et al., 2018).

ROCK MAGNETIC RESULTS

The results of typical rock magnetic measurements are shown in Figure 4.

(1) For the rhyolite and andesite samples from the upper part of section A–A′ (Fig. 4A), the IRM acquisition curve increases rapidly in the lower magnetization field (usually less than 200 mT) but does not reach saturation at 2500 mT, suggesting the coexistence of both high- and low-coercivity magnetic minerals. The IRM component analysis (Kruiver et al., 2001) displays two components, with low-coercivity component 1 contributing most of the saturation IRM. The thermal demagnetization of the composite IRM curves shows that the low- and medium-coercivity components unblock at ~580 °C, and the high- and part of the medium-coercivity components unblock at ~680 °C. These results indicate that magnetite and some hematite coexist in such samples.

(2) For the tuff sample from the middle part of section A–A′ (Fig. 4B), the IRM acquisition curve quickly reaches quasi-saturation at a lower magnetization field (usually less than 200 mT). The IRM component analysis displays an absolute dominance of the low-coercivity magnetic component. The thermal demagnetization of the composite IRM curves show that low- and medium-coercivity components are dominant, and both of them unblock at ~580 °C. These results indicate that magnetite is the main magnetic carrier in the tuff samples.

(3) For the basalt and andesite samples from the lower part of section A–A′ (Fig. 4C), the IRM acquisition curve increases gradually with the increase of the magnetic field and does not reach saturation by 2500 mT. The IRM component analysis displays an absolute dominance of the high-coercivity magnetic component. The thermal demagnetization of the composite IRM curves show that the medium- and high-coercivity components are dominant with an unblocking temperature of ~680 °C. These results indicate that hematite is the primary magnetic carrier in such samples.

(4) For the limestone of section B–B′ (Fig. 4D), the IRM acquisition curve climbs rapidly in the initial low fields of 200–300 mT. After that, it rises steeply and does not reach saturation by 2500 mT, indicating the presence of both low- and high-coercivity minerals. The IRM component analysis displays two-humped components. The thermal demagnetization of the composite IRM curves show that the low- and medium-coercivity components unblock at ~580 °C. Notably, the high-coercivity components show a rapid drop during the initial stages of thermal demagnetization and finally unblock at ~670 °C. These results indicate that magnetite with hematite/goethite is the magnetic carrier in such samples.

The AMS measurements were carried out on samples from section A–A′ to identify the degree of deformation for the studied area (Fig. 5). For the tuff samples, the minimum anisotropy axis is nearly perpendicular to the bedding plane. The maximum and intermediate anisotropy axes parallel the bedding plane with respective dominant directions (Fig. 5A). These features suggest that the samples may belong to weak deformed magnetic or sedimentary fabric related to paleocurrent or formation on a slope (Hrouda and Chadima, 2020). The principal anisotropy directions disperse for the volcanic samples (rhyolite, basalt, and andesite), with some slightly dominant trends (Fig. 5B). The values of modified AMS degree (P_j), magnetic foliation (F), and magnetic lineation (L) are relatively small for both tuff and volcanic samples; the AMS ellipsoid shows both prolate and oblate ellipsoids, and the anisotropy shape factor (T) is independent of P_j. Based on these analyses, we speculated that the rocks in the sampling area may have undergone compression with the deformation of the "pencil structure" to the "weak cleavage" stage (Hrouda and Chadima, 2020). However, they did not subject to intense deformation to affect paleomagnetic studies.

PALEOMAGNETIC RESULTS

The Upper Permian Nayixiong Formation (Section A–A′)

Three hundred and four specimens from section A–A′ underwent stepwise thermal demagnetization. Typical demagnetization diagrams are shown in Figure 6.

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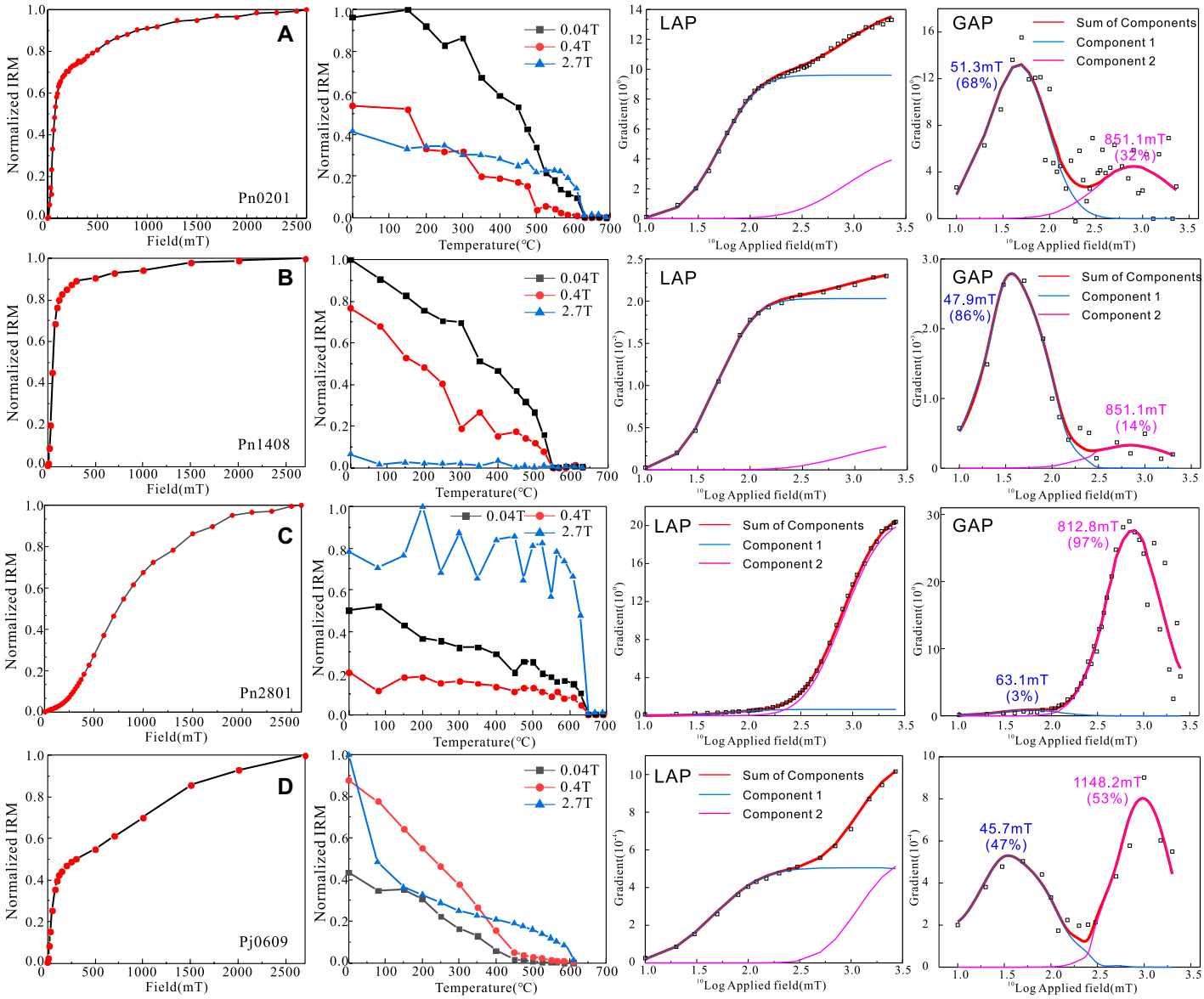


Figure 4. Rock magnetic measurements for representative samples for this study in eastern Tibet. Each case includes the isothermal remanent magnetization (IRM) acquisition curves, the thermal demagnetization of the composite IRM, and the component analysis of coercivity distributions (Kruiver et al., 2001). GAP—gradient acquisition plot; LAP—linear acquisition plot.

The remanence intensity of rhyolite and andesite samples in the upper part of section A–A′ decreases slowly in the NRM-400 °C. It demagnetizes in the range of 450–585 °C or 450–650 °C, consistent with magnetite and some hematite coexisting in these samples as revealed by rock magnetic experiments. The demagnetization curves exhibit two distinctive components. We identify the low-temperature component (LTC) at NRM-400 °C and the ChRM at 450–585 °C or 450–650 °C using the PCA (Figs. 6A and 6B).

Tuff samples in the middle part of section A–A′ exhibit two demagnetization behavior types (Figs. 6C and 6D). The first type shows two-component characteristics; its remanence intensity decreases slowly up to 400 °C, reserving ~50% of NRM at 400 °C, and demagnetizes at 550–565 °C (Fig. 6C). We identify the LTC at NRM-400 °C and the ChRM at 500–565 °C with the PCA method. The remanence intensity for the second type decreases rapidly with increasing temperature to ~20% of NRM at 400 °C and usually below 10% of NRM at 500 °C (Fig. 6D). We obtain the LTC at NRM-400 °C. The ChRM direction exhibits a good motion along the great circle on the stereographic projection plot after 400 °C. Therefore, although some studies have suggested that the GC fit is somewhat less accurate than the PCA (Meng et al., 2012), this method was used for such samples.

The remanence intensity of basalt and andesite samples in the lower part of section A–A′ usually changes steadily or slowly before 500 °C, then decreases gradually after 500 °C and unblocks at 610–650 °C or 610–690 °C (Figs. 6E–6H), consistent with hematite being the primary magnetic carrier as revealed by rock magnetic experiments. The samples exhibit two types: single-component (Figs. 6E and 6F), which stably tends to the distant point from NRM to 690 °C; we identify the ChRM at NRM-690 °C, the other two components (Figs. 6G and 6H). We identify the LTC at NRM-450 °C and

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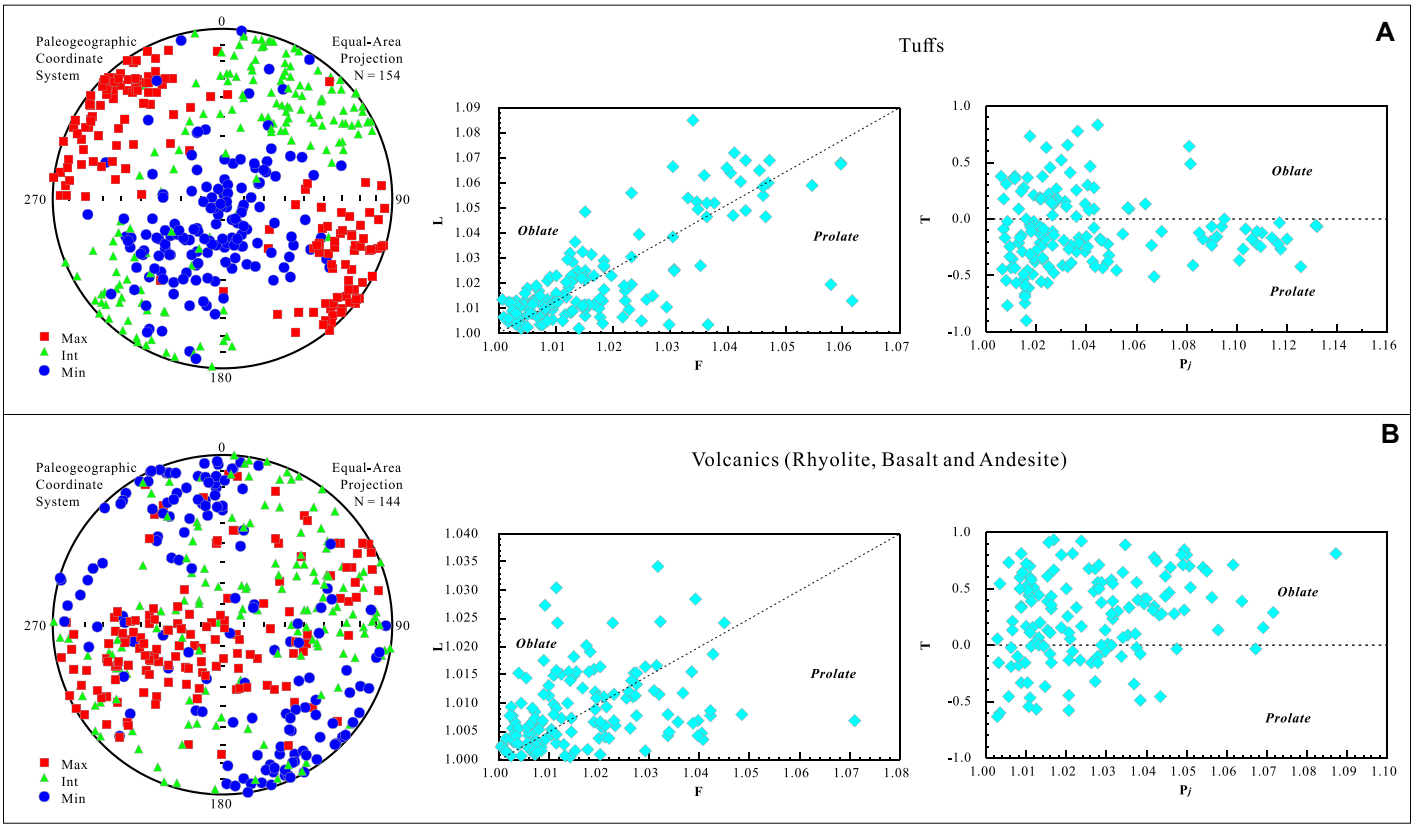


Figure 5. Results of anisotropy of magnetic susceptibility measurements for samples from this study in eastern Tibet: (A) tuffs; (B) volcanics (rhyolite, basalt, and andesite). Each case includes the equal-area stereoplot of principal susceptibility axes in a paleogeographic coordinate system, the plots of magnetic foliation (F) versus magnetic lineation (L), the plots of anisotropy degree (P*j*) versus shape parameter (T). Max—maximum principal axes; Int—intermediate principal axes; Min—minimum principal axes; N—number of samples.

the ChRM at 500–650 °C or 500–690 °C with the PCA method.

We obtained the LTC from 226 samples. The sample-mean directions are: $D_g = 355.6°$, $I_g = 42.6°$, $k_g = 29.5$, $\alpha_{95} = 1.8°$ in geographic coordinates and $D_s = 3.5°$, $I_s = 19.4°$, $k_s = 22.5$, $\alpha_{95} = 2.0°$ in stratigraphic coordinates (Fig. 7A). The mean direction in the geographic coordinates is close to the present-day geomagnetic field (PDF) direction ($D_{PDF} = 359°$, $I_{PDF} = 48°$). Therefore, we interpret it as most likely the viscous remanence direction.

We obtained the ChRMs from all 39 sampling sites in section A–A′ (Table 1; Table S1[1]). We exclude sites pn01 and pn23 from the subsequent site-mean calculation: the mean direction of site pn01 (rhyolite, 9 samples) is quite different from other sites, which is most likely influenced by its flow structure; the site pn23 (welded breccia, 8 samples) is for the dispersion test described in the next section. The sites pn03, pn08, pn09, and pn12 were also excluded from calculating the mean direction because their cones of confidence α_{95} are larger than 15° or no ChRM could be obtained using the PCA. After that, we use 33 sites (248 samples) to calculate the site-mean direction as: $D_g = 266.4°$, $I_g = 5.7°$, $k_g = 43.7$, $\alpha_{95} = 3.8°$ in geographic coordinates, and $D_s = 271.4°$, $I_s = 17.5°$, $k_s = 42.9$, $\alpha_{95} = 3.9°$ in stratigraphic coordinates (Fig. 7B). We used the site-dependent paleosecular variation (PSV) test of Deenen et al. (2011) to explore if the virtual geomagnetic poles (VGPs) average the secular variation. The 33 VGPs yield an $A_{95} = 3.2°$, which falls into the N-dependent envelope of $A_{95max} = 9.06°$ and $A_{95min} = 2.96°$, indicating that PSV is averaged in our study.

The Middle Permian Jiushidaoban Formation (Section B–B′)

We measured samples from seven new sites from section B–B′ where samples from only five were published by Cheng et al. (2013) and reanalyzed samples from all 12 sites to obtain a statistically improved result for the Middle Permian limestones (P_2j). The NRM intensity for these samples is ~1.23 mA/m. The limestone samples from the published five sites used step-by-step thermal or AF demagnetization methods (Figs. 8A and 8B). Considering the rock-magnetic experiments revealed the presence of high-coercive magnetic minerals such as goethite in the limestone, we subjected most of the samples from the seven new sites to the hybrid demagnetization procedure. The results show two remanent magnetization components for most samples (Figs. 8C–8F). We identify the LTC at NRM-250 °C and the high field ChRM between 20 mT and 100 mT.

We obtained the LTC from 59 samples. The sample-mean directions are $D_g = 350.0°$, $I_g = 55.7°$, $k_g = 28.1$, $\alpha_{95} = 3.6°$ in situ and $D_s = 295.1°$, $I_s = 36.6°$, $k_s = 9.2$, $\alpha_{95} = 6.5°$ after tilt correction, which is close to the PDF direction in geographic coordinates. We interpret this component as a remagnetization by the PDF (Fig. 9A).

The results of the combined newly obtained seven site-mean directions of the ChRM and pre-

[1]Supplemental Material. Data used in this study (Tables S1–S3). Please visit https://doi.org/10.1130/GSAB.S.22357690 to access the supplemental material, and contact editing@geosociety.org with any questions.

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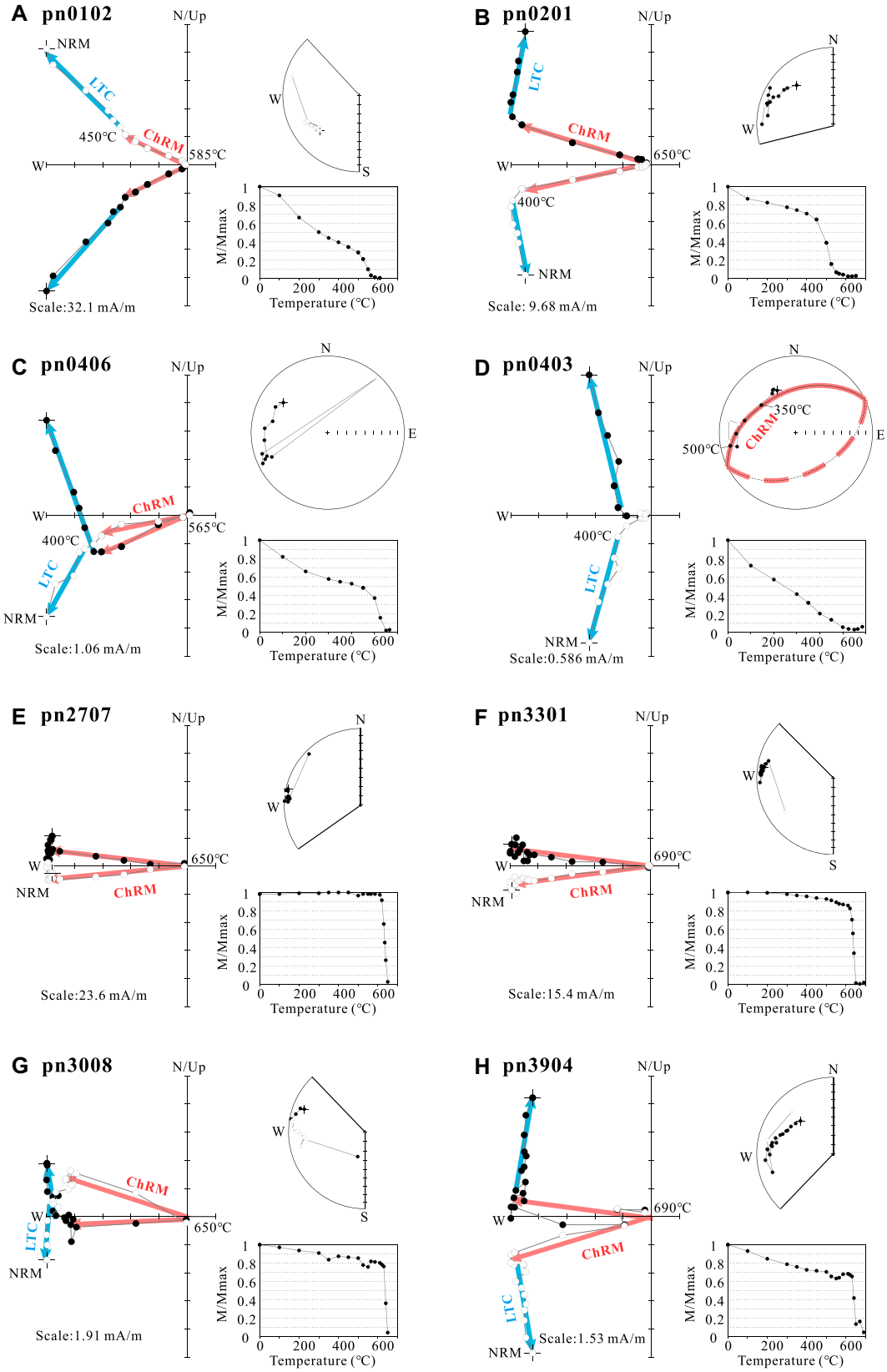


Figure 6. Representative demagnetization diagrams of Upper Permian volcanic samples in geographic coordinates from this study in eastern Tibet. Solid/open circles in orthogonal plots represent projections to the horizontal/vertical plane, and solid/open circles in stereoplots represent projections to the lower/upper hemisphere. The thick blue and pink lines represent the fitted directions corresponding to the low-temperature component (LTC) and the characteristic remanence component (ChRM). M/Mmax—magnetization/maximum magnetization; NRM—natural remanent magnetization.

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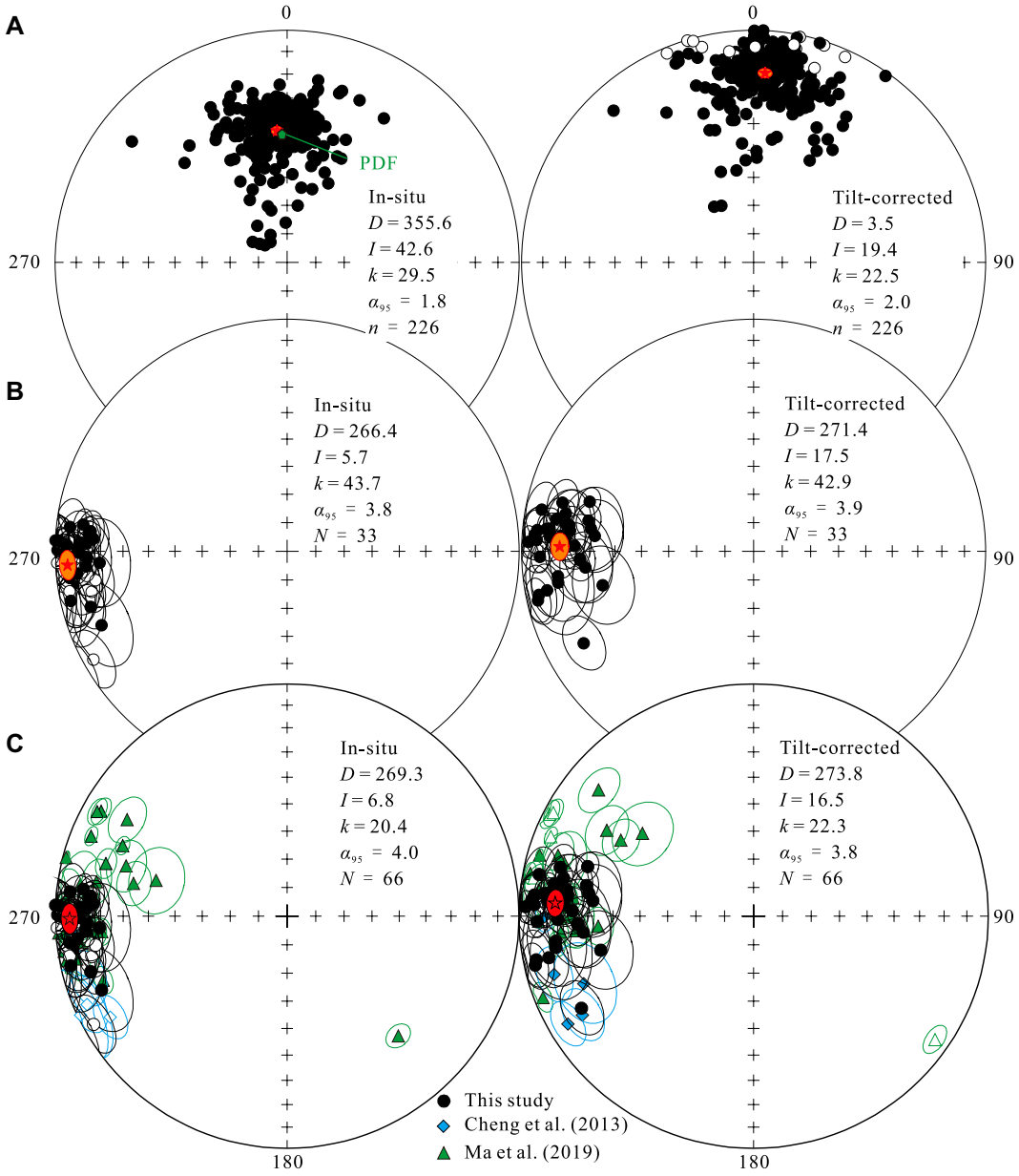


Figure 7. Equal-area projections of the components isolated from the Upper Permian samples, from this study in eastern Tibet, before (left) and after (right) tilt correction. (A) Low-temperature component; (B) site-mean directions of the characteristic remanence component from this study; (C) mean direction calculated by using the sites from Cheng et al. (2013) and Ma et al. (2019) and this study. Open/closed circles are projections on the upper/lower hemisphere. PDF—present-day geomagnetic field direction; *D*—declination; *I*—inclination; *k*—best estimate of the precision parameter; α95—radius of the mean direction at the 95% confidence level; n—number of samples used in mean calculation; N—number of sites used in mean calculation.

viously published five site-mean directions are $D_g = 336.5°$, $I_g = 74.3°$, $k_g = 14.9$, $\alpha_{95} = 11.6°$ in situ and $D_s = 265.0°$, $I_s = 42.6°$, $k_s = 24.6$, $\alpha_{95} = 8.9°$ after tilt correction (Fig. 9B, Table 1; Table S1).

DISCUSSION

Origin of the Characteristic Remanence

To prove the primary character of the ChRM, we performed a few field tests. Section A–A′ (P_3n) and section B–B′ (P_2j) are from the two wings of the Zhouqiongmalu compound anticline with opposite tendencies that make them perfect for the fold tests. The site-mean combined direction for the Middle and Upper Permian ChRMs before and after tilt correction are $D_g = 271.3°$, $I_g = 22.9°$, $k_g = 5.0$, $\alpha_{95} = 10.7°$ and $D_s = 270.0°$, $I_s = 24.1°$, $k_s = 21.6$, $\alpha_{95} = 4.7°$, $N = 45$ sites, respectively (Fig. 10A; Table 1). The McElhinny's (1964) fold test is positive: $N = 45$ sites, $k_s/k_g = 23.1/4.8 = 4.34 > F(2*(n2–1), 2*(n1–1)) = 1.65$, at the 99% confidence level. McFadden's (1990) fold test is also positive: $N = 45$ sites, in situ $\xi_2 = 33.73$, tilt-corrected $\xi_1 = 3.833$, statistical threshold $\xi = 11.03$ at the 99% confidence level. The Watson and Enkin's (1993) fold test provides an optimal concentration at 120.1% ± 4.4% unfolding (Fig. 10B). We note that the unfolding test value is much higher than 100%, which may be because the site-mean inclination of the Upper Permian ChRM is ~20° lower than that of the Middle Permian ChRM. It is probably a response to the northward drift of the NQB from the Middle Permian to the Late Permian (see next section). As mentioned earlier, the Zhouqiongmalu compound anticline was deformed at least before the Jurassic and was probably formed before the Late Triassic because the Upper Triassic strata covered the late Paleozoic strata with an angular unconformity in the study area. The positive fold tests constrain the timing of the Middle and Upper Permian ChRMs acquisition as the pre-Late Triassic.

We drilled eight samples (pn23 site) from the welded breccia in section A–A′ for a disper-

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TABLE 1. PALEOMAGNETISM RESULTS FROM SECTION A–A′ AND SECTION B–B′, TIBET

Site ID	Lithology	Bedding Strike/dip	n(n_0)/N	In-situ D_g (°)	In-situ I_g (°)	Tilt-corrected D_s (°)	Tilt-corrected I_s (°)	k	α_{95}
Section A–A′, Upper Permian Nayixiong Formation (P_3n), latitude: 33°52′17.0″, longitude: 91°55′04.4″									
Pn01*	Rhyolite	317/20	9/9	265.2	–16.1	263.5	–0.1	7.3	20.5
Pn02	Andesite	321/19	10/10	274.9	15.3	280.5	28.4	102.2	4.8
Pn03*	Tuff	322/21.5	5/5	265.6	28.2	275.8	45.1	14.7	20.6
Pn04	Tuff	324/24	8(4)/8	261.0	6.9	264.5	28.0	27.7	10.7
Pn05	Tuff	327.5/31	10(7)/10	267.7	–9.0	269.0	17.8	52.9	6.7
Pn06	Tuff	302/30.5	6(3)/7	267.6	3.6	273.3	19.9	67.7	8.2
Pn07	Tuff	322/23.5	5(2)/6	261.6	–4.3	262.7	16.1	37.2	12.7
Pn08*	Tuff	317/25	5(3)/5	266.1	17.3	274.8	35.6	5.4	36.3
Pn09*	Tuff	317/25	4(4)/4	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.
Pn10	Tuff	317/25	5(3)/6	263.0	6.3	267.6	26.1	35.1	13.1
Pn11	Tuff	296/27	7(6)/7	248.3	15.0	257.3	33.7	25.4	12.2
Pn12*	Tuff	296/27	5(3)/7	283.9	22.7	296.2	25.7	14.9	20.5
Pn13	Tuff	293/28	7(5)/9	270.8	10.6	278.1	19.6	36.7	10.1
Pn14	Tuff	294/26	8(5)/8	266.5	20.1	277.9	29.9	20.0	12.7
Pn15	Tuff	312/29	6(5)/7	277.5	12.4	286.8	27.1	243.6	4.3
Pn16	Tuff	312/24	6(5)/8	240.8	–4.9	241.6	17.8	67.7	8.2
Pn17	Tuff	317/24	9(5)/9	267.3	6.0	271.7	23.8	58.3	6.8
Pn18	Tuff	317/25	7(2)/7	259.6	–5.2	260.9	15.8	17.6	14.8
Pn19	Tuff	317/23.5	8/8	258.3	–15.1	257.2	5.2	92.1	5.8
Pn20	Tuff	297/25	10(5)/11	269.6	11.1	276.4	21.4	47.2	7.1
Pn21	Tuff	277/34	8(6)/8	273.7	17.4	284.3	16.1	71.4	6.6
Pn22	Tuff	306/36	9(5)/11	263.9	9.9	275.5	31.8	22.5	11.1
Pn23*	Welded Breccia	292/50	0/8	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.
Pn24	Basalt	292/50	7/7	267.1	–0.9	274.6	18.2	117.1	5.6
Pn25	Basalt	280.5/37	6/10	270.3	1.2	273.0	7.1	61.6	8.6
Pn26	Basalt	280.5/37	6/7	254.2	13.0	267.1	26.0	117.7	6.2
Pn27	Basalt	280.5/37	7/8	272.1	–2.4	272.4	3.2	217.6	4.1
Pn28	Basalt	280.5/37	8/8	276.4	6.5	281.1	7.7	57.0	7.4
Pn29	Basalt	280.5/37	5/5	272.4	2.7	275.6	7.0	153.2	6.2
Pn30	Basalt	299/37	7/8	261.4	–17.6	258.5	6.2	60.8	7.8
Pn31	Basalt	284/31	8/8	272.1	14.5	281.4	18.6	213.3	3.8
Pn32	Basalt	284/31	10/11	268.7	6.3	274.0	13.3	54.5	6.6
Pn33	Basalt	284/31	10/10	271.6	5.6	276.2	11.2	171.3	3.7
Pn34	Basalt	284/31	6/6	272.7	0.1	274.3	5.9	107.2	6.5
Pn35	Basalt	276.5/17.5	7/8	272.5	11.8	276.3	12.4	113.1	5.7
Pn36	Basalt	246/15	10/10	265.7	14.1	268.5	8.6	133.2	4.2
Pn37	Basalt	274/13	7/7	275.8	16.0	279.4	15.1	109.2	5.8
Pn38	Basalt	276.5/17.5	8/8	257.1	4.8	259.3	10.3	45.5	8.3
Pn39	Basalt	284/31	7/7	267.9	14.8	278	20.9	34.6	10.4
Formation mean by specimens			248(68)/306	267.1	6.4	N.D.	N.D.	30.2	1.6
			N.D.	N.D.	N.D.	271.3	17.5	29.6	1.7
Formation mean by sites			33/39	266.4	5.7	N.D.	N.D.	43.7	3.8
			N.D.	N.D.	N.D.	271.4	17.5	42.9	3.9
Mean Pn†			66	269.3	6.8	273.8	16.5	22.3	3.8
Section B–B′, Middle Permian Jlushidaoban Formation (P_2j), latitude: 33°57′41.6″, longitude: 91°48′37.1″									
Pj01§	Limestone	124/40	5/5	259.4	77.5	225.6	40.6	36.1	12.9
Pj02§	Limestone	142/62	12(2)/14	7.3	64.3	257.0	44.0	14.7	11.7
Pj03§	Limestone	142/67	9(2)/12	334.8	57.0	268.2	26.1	23.3	10.9
Pj04§	Limestone	131/71	7(1)/9	341.8	51.2	261.4	33.8	27.1	11.8
Pj05§	Limestone	140/71	9(5)/10	332.2	51.3	271.4	22.3	26.9	10.1
Pj06	Limestone	140/62	10/10	5.0	66.5	252.5	42.8	90.7	5.1
Pj07	Limestone	151/64 & 140/64	6/8	354.4	56.0	273.6	37.5	45.8	10.0
Pj08	Limestone	204/40.5	8/10	182.1	78.5	276.3	52.5	69.3	6.7
Pj09	Limestone	161/45	9/10	21.9	71.8	275.2	54.8	50.7	7.3
Pj10	Limestone	161/53	9/11	46.1	84.5	254.1	41.9	38.5	8.4
Pj11	Limestone	169/31	4/5	342.8	64.6	299.3	48.6	41.1	14.5
Pj12	Limestone	169/38	11/11	354.2	82.3	271.5	52.1	16.7	11.5
Formation mean by specimens			99(10)/115	350.3	72.3	N.D.	N.D.	14.5	3.9
			N.D.	N.D.	N.D.	264.6	42.8	16.4	3.6
Formation mean by sites			12/12	336.5	74.3	N.D.	N.D.	14.9	11.6
			N.D.	N.D.	N.D.	265.0	42.6	24.6	8.9

Notes: The characteristic remanent magnetization (ChRM) statistics of the two sections are calculated together for generalizing fold tests. The results show that the average directions of ChRM before and after tilt correction are $D_g = 271.3°$, $I_g = 22.9°$, $k = 5.0$, $\alpha_{95} = 10.7$, and $D_s = 270.0°$, $I_s = 24.1°$, $k = 21.6$, $\alpha_{95} = 4.7$, and n = 45. Fold tests: (1) Watson and Enkin's (1993) test: the optimum concentration (maximum k) is achieved at 120.1% ± 4.4% unfolding; (2) McFadden's (1990) test: n = 45 sites, in-situ $\xi 2 = 33.73$, tilt-corrected $\xi 1 = 3.833$, statistical threshold $\xi = 11.03$ at 99% confidence level, indicative of a positive fold test; (3) McElhiny's fold test (1964): N = 45, $k_s/k_g = 23.1/4.8 = 4.34 > F$ (2*(n2 − 1), 2*(n1 − 1)) = 1.65, at the 99% confidence level fold test is positive. Site ID—site identification; Strike/dip—strike azimuth and dip of bed; N.D.—no data. n(n_0)/N = number of samples or sites used in mean calculation/demagnetized; number in parentheses indicates number of remagnetization circles used; D_g, I_g (D_s, I_s), declination and inclination of direction in-situ (after tilt correction); k and α_{95}, the best estimate of the precision parameter and the radius of the mean direction at the 95% confidence level, respectively.

*Site-mean directions discarded from final formation mean calculation.

†Mean Pn represents a formation mean direction calculated using 33 sites in this paper, 28 sites in Ma et al. (2019), and five sites in Cheng et al. (2013).

§Site-mean directions from Cheng et al. (2013).

sion test. For these samples, stable ChRM were obtained at ~500–650 °C or 500–670 °C using the PCA method (Fig. 10C), whereas the dispersion of the eight ChRMs is extensive (Fig. 10D). Watson's (1956) random distribution test shows that the distribution is indeed random at the 99% and 95% probability levels (R = 3.63, R_0 at 95% = 4.48, R_0 at 99% = 5.26). We did not have the possibility to perform a baking test like the method suggested by Meng et al. (2022). The positive fold and dispersion tests suggest that the statistically significant ChRMs separated from most basalt, andesite, and tuff sites in section A–A′ are most likely a primary remanent magnetization.

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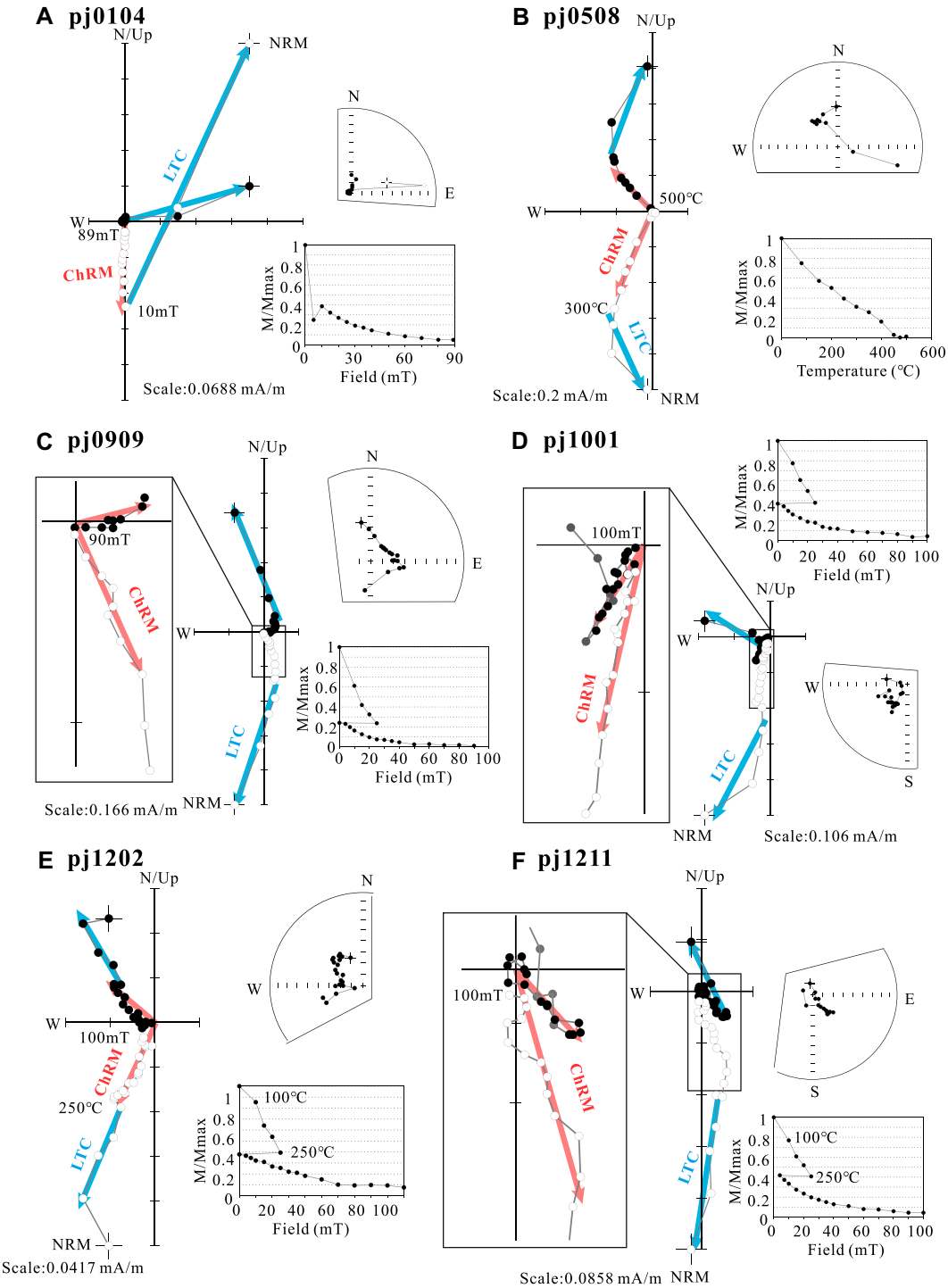


Figure 8. Representative demagnetization diagrams of Middle Permian limestones from this study in eastern Tibet, in geographic coordinates. Symbols are the same as in Figure 6.

We infer the primary character of the ChRM from section B–B′ (P_2j, limestones) from the following observations. First, the fold tests demonstrated above are positive. Second, the position of the paleopole is distinct from that of all published paleopoles of the NQB since the Late Permian (e.g., Cheng et al., 2012a; Song et al., 2015, 2020; Yan et al., 2016; Zhou et al., 2019). Third, the paleopole corresponds to a paleolatitude of 24.7° for the NQB, which is consistent with new paleontological discovery and paleoclimate evidence that suggest the NQB located at mid-latitudes in the southern hemisphere during the Early and Middle Permian (Zhang et al., 2016; Qiao et al., 2021; Zhang and Torsvik, 2022). Finally, our paleomagnetic data show that the NQB drifted northward from the Middle Permian to the Late Permian, which is consistent with the unidirectional northward drift pattern of the Tethys blocks (Wu et al., 2020; Zhu et al., 2022).

The Late Paleozoic Paleogeography and Drift History of the North Qiangtang Block

The NQB is located in the northern part of the present Qinghai-Tibet Plateau, with an aver-

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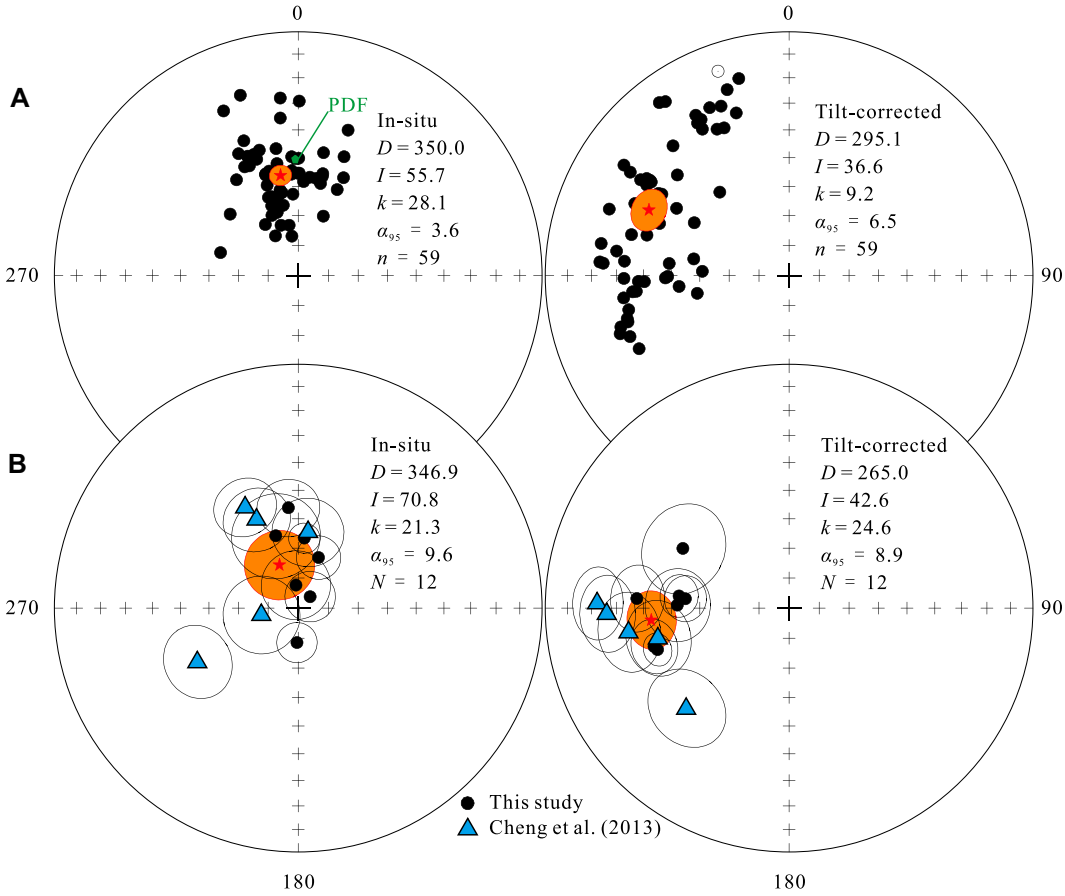


Figure 9. Equal-area projections of components isolated from the Middle Permian limestones from this study in eastern Tibet, before (left) and after (right) tilt correction. (A) Low-temperature component; (B) site-mean directions include seven newly obtained sites and five sites from Cheng et al. (2013). Symbols are the same as in Figure 7.

age elevation of more than 5000 m. In order to reconstruct the paleogeography of the NQB in geological history, a series of meaningful paleomagnetic studies have been carried out in the Qiangtang area in the last two decades. Table S2 shows the Silurian–Triassic paleomagnetic data collected from this block and corresponding references. These data are from the Tanggula, Raggyorcaka, Qamdo-Zadoi, and Longmuco areas. We plot these poles on an equal-area projection of the world (Fig. 11).

Paleomagnetic poles in the Tanggula area correspond to successive geological periods of the Late Carboniferous, Early Permian, Middle Permian, Late Permian, Middle Triassic, and Late Triassic. Among them, the Late Carboniferous paleomagnetic pole from the limestone of C_2z has a positive conglomerate test (Yang et al., 2017). The Early Permian pole from the volcanic rocks of the P_1nr passes the fold test (Song et al., 2017). This paper contributes to the Middle Permian pole from the limestone of the P_2j, which includes five sites of Cheng et al. (2013). The Late Permian pole is derived from statistics of paleomagnetic data of the P_3n volcanic rocks. Cheng et al. (2013) and Ma et al. (2019) reported paleomagnetic data from the P_3n, and these two data sets are consistent with our results. Therefore, we count these three data sets (including 66 site-mean directions) together to generate a new robust Late Permian pole. The results are $D_g = 269.3°$, $I_g = 6.8°$, $k_g = 20.4$, $\alpha_{95} = 4.0°$ in situ and $D_s = 273.8°$, $I_s = 16.5°$, $k_s = 22.3$, $\alpha_{95} = 3.8°$ after tilt correction, which yields a paleomagnetic pole at 7.8°N, 6.9°E, $A_{95} = 2.8°$ (Fig. 7C; Table 1; Table S2). Song et al. (2020) reported the paleomagnetic data from the Middle Triassic (242–240 Ma) lavas. There are two sets of paleomagnetic data in the Late Triassic from volcanic rocks (Song et al., 2015) and sedimentary rocks with the inclination flattening correction (Zhou et al., 2019). Both data sets have positive fold and reversal tests.

There are six paleomagnetic poles in the Raggyorcaka area (Cheng et al., 2012b; Song et al., 2012; Zhou et al., 2019). The Permian paleomagnetic data successfully pass the reversal, fold, and baked contact tests, indicating that these should be primary remanence (Cheng et al., 2012b). We note discrepancies in the Late Carboniferous and Early Permian NQB paleolatitudes calculated from paleomagnetic data of the Tanggula and the Raggyorcaka areas (Fig. 11; Table S2). Most likely, the small number of samples in the study caused considerable uncertainty in the data from the Raggyorcaka area. Another possibility is a slight relative movement and adjustment between these two areas during the Late Carboniferous–Middle Permian.

There are four paleomagnetic poles in the Qamdo-Zadoi area. The positions of the two Late Triassic paleomagnetic poles (Lin and Watts, 1988; Yu et al., 2022) are close to the positions of contemporaneous poles from the Tanggula area. However, the positions of the Late Permian (Huang et al., 1992) and Late Permian–Early Triassic (Guan et al., 2021) paleomagnetic poles differ significantly from those of the same geological age in the Tanggula and the Raggyorcaka areas (Fig. 11; Table S2). The significance of this remains to be clarified.

We use more successive and reliable paleomagnetic data from the Tanggula area to represent the NQB for further interpretation (Fig. 12; Table S3). We note that most of the Carboniferous and Permian paleomagnetic data show a declination to the southwest or close to the west with a downward inclination similar to the Middle and Late Permian data obtained in this study. Meanwhile, some of the Carboniferous (Yang et al., 2017) and Early Permian (Song et al., 2017) paleomagnetic data show declination to the northeast with upward inclination, which is nearly opposite in both types of direc-

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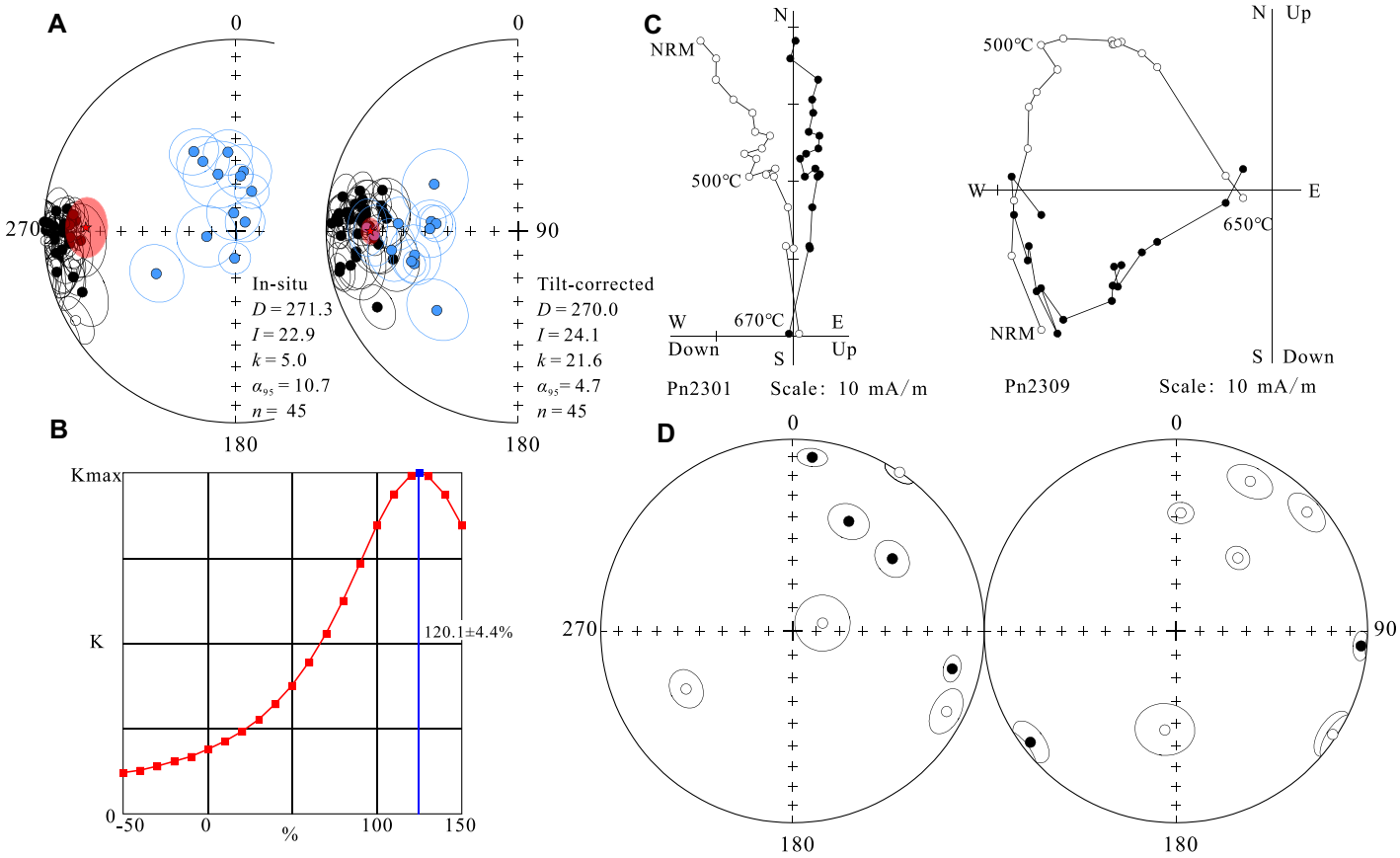


Figure 10. (A) Equal-area projections of the site-mean direction for all the 45 sites from the Upper Permian (section A–A′) and the Middle Permian (section B–B′) rocks from this study in eastern Tibet, before (left) and after (right) tilt correction. (B) The incremental unfolding analysis result (Watson and Enkin, 1993). (C) Orthogonal plots for representative samples of the welded breccia (site Pn23) in geographic coordinates. Solid/open symbols represent projection onto the horizontal/vertical plane. (D) Equal-area projections of the characteristic remanence components isolated from the eight welded breccia samples. Open/closed circles are projections on the upper/lower hemisphere. NRM—natural remanent magnetization; K—precision parameter; *D*—declination; *I*—inclination; *k*—best estimate of the precision parameter; α_{95}—radius of the mean direction at the 95% confidence level; n—number of samples used in mean calculation.

tions in the same time interval, predicting that the two sets of directions were obtained in different polar periods. New paleontological discoveries and paleoclimate evidence suggest the NQB was located in the southern hemisphere midlatitudes during the Early and Middle Permian (Zhang et al., 2016; Qiao et al., 2021; Zhang and Torsvik, 2022). Therefore, we interpret the directions of the west and southwest declination with a downward inclination as remanence acquired in the interval of prevailed reverse polarity and the directions of the northeast declination with an upward inclination as remanence acquired in the interval of normal polarity. The corresponding NQB paleolatitudinal evolution matches the suggestion of Wu et al. (2020) and Zhu et al. (2022) that the Tethys blocks drifted northward unidirectionally (see next paragraph). We also note that the Carboniferous and Permian poles are from limestones (Carboniferous and Middle Permian) and volcanic rocks (Early Permian and Late Permian). Although some researchers suggest that the limestones should be corrected for the inclination flattening (Maron and Muttoni, 2021), the perfect consistency between the paleomagnetic poles from the Early Permian igneous and the Late Carboniferous, Middle Permian limestones suggest that the influence of flattening correction in those limestones is negligible. Therefore, we do not apply any inclination flattening correction to those data.

The paleomagnetic data discussed above elucidate the NQB drift history from the Late Carboniferous to Late Triassic. The NQB was stably located at ~24°S from the Late Carboniferous to Middle Permian (reference point 34°N, 92°E chosen in the middle of the block) (Fig. 12; Table S3). The stable paleolatitude position of the NQB may have persisted from the Silurian to the Middle Permian, as newly reported Early–Middle Silurian paleomagnetic data from the Longmuco area show a similar paleolatitude of 23.9°S ± 9.7° (Jiang et al., 2022). The NQB drifted rapidly northward after the Middle Permian, reaching ~8.4°S in the Late Permian, ~23.5°N in the Middle Triassic, and ~32.9°N in the Late Triassic. The average plate motion velocity from the Middle Permian to the Late Permian is ~16.4 cm/yr, comparable to the highest velocities of plate motion known today (DeMets et al., 2010).

To further determine the paleogeographic position of the NQB in Tethys, we collated the Early Permian to Triassic paleomagnetic data from the Tarim, North China Block (NCB) (Huang et al., 2018), South China Block (SCB) (Huang et al., 2018), India of Gondwana (Torsvik et al., 2012), SQB (Wei et al., 2022; Song et al., 2012), Sibumasu (Huang and Opdyke, 1991; Zhao et al., 2015, 2020), Lhasa (Ran et al., 2012; Zhou et al., 2016), and Indochina Block (IC) (Huang and Opdyke, 2016; Chi et al., 2016; Yan et al., 2017, 2019, 2020; Jia

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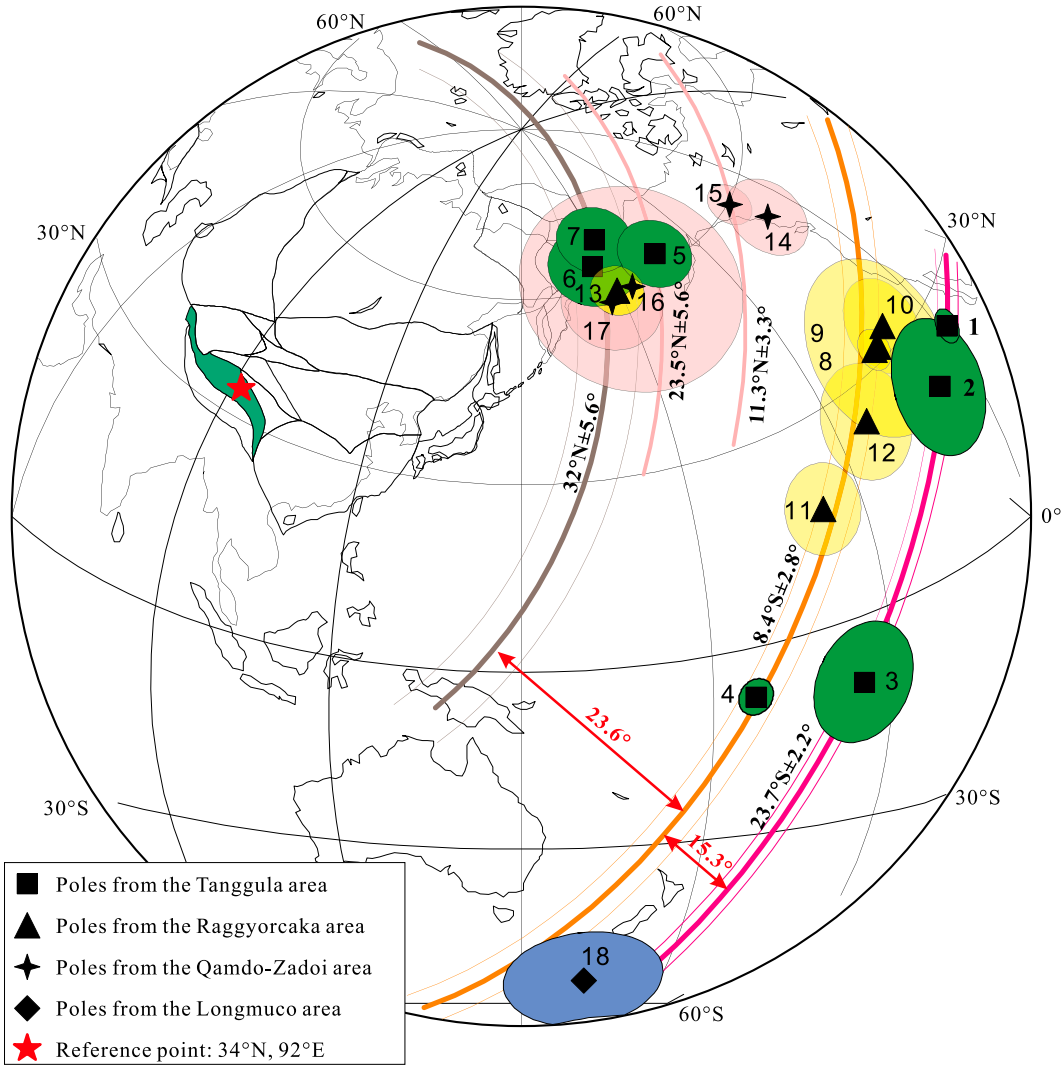


Figure 11. Equal-area projection to illustrate the Silurian–Triassic paleomagnetic poles from the North Qiangtang Block of the present Tibetan Plateau. 1—Late Carboniferous (C_2), Yang et al. (2017); 2—Early Permian (P_1), Song et al. (2017); 3—Middle Permian (Mean P_2), calculated from Cheng et al. (2013) and this study; 4—Late Permian (Mean P_3), calculated from Cheng et al. (2013), Ma et al. (2019), and this study; 5—Middle Triassic (T_2), Song et al. (2020); 6—Late Triassic (T_3), Song et al. (2015); 7—Late Triassic (T_3), Zhou et al. (2019); 8—Late Carboniferous (C_2), Cheng et al. (2012b); 9—Early–Middle Permian ($P_{1–2}$), Cheng et al. (2012b); 10—Late Permian (P_3), Cheng et al. (2012b); 11—Early Triassic (T_1), Cheng et al. (2012b); 12—Early Triassic (T_1), Zhou et al. (2019); 13—Late Triassic (T_3), Song et al. (2012); 14—Late Permian (P_3), Huang et al. (1992); 15—Early Triassic (T_1), Guan et al. (2021); 16—Late Triassic (T_3), Lin and Watts (1988); 17—Late Triassic (T_3), Yu et al. (2022); 18—Early–Middle Silurian ($S_{1–2}$), Jiang et al. (2022).

et al., 2022), and calculated their corresponding paleolatitudes with a common reference point to the NQB (Fig. 12; Table S3). Figure 12 demonstrates that the NQB and IC likely shared similar paleolatitudinal and kinematic processes during the late Paleozoic–early Mesozoic. This result corroborates the hypothesis that the late Paleozoic NQB and IC were geographically linked, as Yan et al. (2019) proposed. These paleomagnetic data further reveal that during the late Paleozoic, at least during the Late Carboniferous to Middle Permian, the NQB and IC were stably located at mid-latitudes (~20°S–30°S) in the southern hemisphere; the Tarim and NCB were located at mid-latitudes (~20°N–35°N) in the Northern Hemisphere, near the southern margin of northern Eurasia (Kravchinsky et al., 2002; Wu et al., 2017; Huang et al., 2018; Wei et al., 2020); the SCB was located near the equator (Wu et al., 2017; Huang et al., 2018); and India—as part of Gondwana—was located at high latitudes (~55°S) in the Southern Hemisphere (Torsvik et al., 2012; Domeier et al., 2021). The NQB and IC likely shared the same metamorphic basement and sedimentary cover (Wu et al., 2020; and references therein); therefore, we suggest that the two blocks may have jointly formed a ribbon-like continental archipelago (named as the NQB-IC continent for convenience in this paper), stably located in the middle of the Paleozoic Paleo-Tethys Ocean.

Implications for the Evolution of the Paleo-Tethys Ocean

The Paleozoic paleogeography and drift history of the NQB provide necessary constraints for the evolution of the Tethys Ocean. First, our study reveals that the LSO as the south branch of the Paleo-Tethys Ocean may have existed between the NQB-IC and the Gondwana continents for a long time in the late Paleozoic. It likely has existed since the Silurian, as the NQB was also located at ~24°S in the Early to Middle Silurian (Jiang et al., 2022). Paleomagnetic data indicate that the latitudinal width of the LSO is ~3000 km, which is smaller than that of the northern branch of the Paleo-Tethys Ocean between the NQB-IC and the southern margin of Laurasia (the remnants of which may now be preserved along the XJS and EKL-ANMQS). The northern branch of the Paleo-Tethys Ocean is ~5000 km (Fig. 12). This interpretation aligns with the new geological evidence (e.g., Li, 1987; Zhang et al., 2006a, 2006b; Zhai et al., 2011; Tang and Zhang, 2014; Zhang et al., 2016; Liu et al., 2019) and existing paleomagnetic models (e.g., Cheng et al., 2012b, 2013; Song et al., 2017; Ma et al., 2019; Zhou et al., 2019).

Second, we suggest that the east Cimmerian blocks, represented by the SQB and Sibumasu (Hennig et al., 2009; Peng et al., 2013), split off the northern margin of the Gondwana con-

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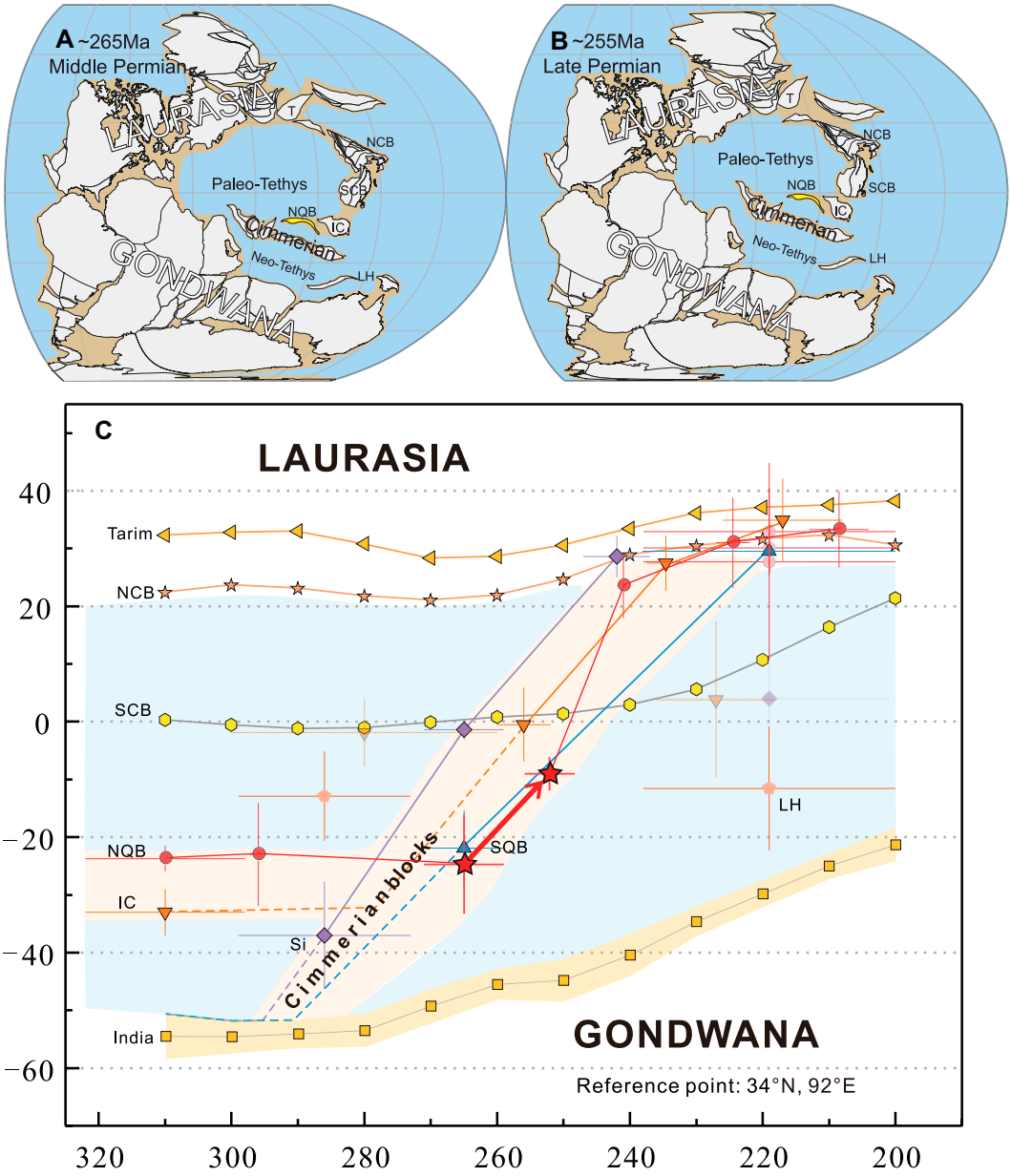


Figure 12. (A and B) Paleogeographic reconstructions of the Tethyan region in the Middle Permian (ca. 265 Ma) and Late Permian (ca. 255 Ma). The locations of the Gondwana, Laurasia, North China Block (NCB), and South China Block (SCB) are according to Huang et al. (2018). The locations of the Cimmerian blocks as a coherent unit are according to Wei et al. (2022). The locations of the North Qiangtang Block–Indochina (NQB-IC) are reconstructed based on this study and Yan et al. (2019, 2020). (C) Paleolatitude evolution of the NQB and its surrounding blocks during the Early Permian to Late Triassic, showing relative positions of the NQB and surrounding blocks, as well as the distribution of the land and sea. Reference point 34°N, 92°E; see Table S3 (see text footnote 1) for detailed paleomagnetic data. SQB—South Qiangtang Block; Si—Sibumasu; LH—Lhasa; T—Tarim.

tinent and drifted northward to ~22°S before the Middle Permian (Wei et al., 2022). These blocks approached or partially collided with the NQB-IC continent (Fig. 12), which likely led to the gradual closure of the LSO and allowed the NQB-IC continent to begin its rapid northward drift. Some geological evidence supports this hypothesis; scholars have found widespread unconformities between the Middle and Late Permian strata in the Qiangtang area (Chen et al., 2006; Qiao et al., 2021), which may be related to the initial collision of the NQB and SQB. Recently published detrital zircon geochronology from the Middle Permian sandstone in the NQB suggests a high similarity in detrital zircon patterns between the NQB and SQB (Zhang et al., 2022), which is consistent with our inference that the two blocks may have merged in the Middle Permian. However, we cannot exclude the alternative possibility that the SCB-IC archipelago laid farther east, as suggested by Zhang and Torsvik (2022). In this case, the LSO may have closed later than Late Permian.

After the Middle Permian, the NQB and SQB drifted rapidly northward, reaching ~9°S in the Late Permian (Fig. 12B). Paleomagnetic data show that the SQB and NQB both reached a latitude of ~32°N in the Late Triassic and collided with the southern margin of northern Eurasia (Song et al., 2012; Song et al., 2017; Zhou et al., 2019; Yu et al., 2022), leading to the final closure of the Paleo-Tethys Ocean.

CONCLUSIONS

Our study reports new Middle and Late Permian paleomagnetic poles of the NQB from the Tanggula area of eastern Tibet. This result, combined with the previously published paleomagnetic data, provides further constraints for the paleogeography and drift history of the NQB and plate tectonic evolution of the Tethys Ocean. The NQB was stably located at the latitude of ~24°S from Late Carboniferous to Middle Permian, likely belonging to a ribbon-like continental archipelago that included the NQB and IC, which lay in the middle of the Paleo-Tethys Ocean during the Paleozoic. As the east Cimmerian blocks, including the SQB, rifted away

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from the northern margin of Gondwana and drifted northward at the beginning of the Early Permian, they reached a latitude of ~22°S in the Middle Permian. They approached the latitude of the NQB-IC continent. The SQB and NQB drifted northward rapidly after the Middle Permian and reached a near-equatorial position in the Late Permian. After that, the SQB and NQB continued to drift northward together and accreted to the southern margin of Eurasia before the Late Triassic. It led to the closure of the Paleo-Tethys Ocean and the formation of the Longmuco–Shuanghu suture between the NQB and SQB.

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